CHAPTER TWO

PHILOSOPHY OF SCIENCE AND THEORY CONSTRUCTION

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Health promotion research, in general, comprises some or all of the nine steps that were delineated in Chapter One and that are undertaken to investigate health-related behaviors. The overarching goal is to understand better ways in which we can influence health behaviors and ultimately health by first identifying the behavioral risk factors for a particular disease. By first understanding and then affecting behaviors that contribute to health, health promotion researchers can have a substantive impact on the associated morbidity and mortality. When conceptualizing and undertaking health promotion research, however, in addition to knowing the research process, it is also important to understand the underlying philosophy.

Derived from the Latin word *scientia*, the literal meaning of *science* is knowledge. Yet, science can also be viewed as a process. The process entails the systematic gathering of seemingly disparate facts and then organizing them into an interconnected whole. Science is the process of acquiring knowledge for the purpose of having knowledge. Why is science important, and why do we hold science in such reverence? Because science is knowledge, and knowledge is the source of empowerment. Creating a body of knowledge regarding myriad subjects such as the properties of matter and energy (physics), the structure of matter (chemistry), living organisms (biology), human behavior (psychology), or

human societies (sociology), to name a few scientific disciplines, can benefit greatly the human experience and provide insight into how nature works and how we as human beings function within nature and with each other. Think of science as an indispensable tool much like the wheel that can take us where we want to go by aiding us in our understanding of, and in our attempt to control, our environment.

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In a Nutshell

Derived from the Latin word scientia, *the literal meaning of* science *is knowledge. Yet, science can also be viewed as a process. The process entails the systematic gathering of seemingly disparate facts and then organizing them into an interconnected whole.*

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Health promotion research also involves a process. In Chapter One, we referred to this as a discovery process. Interestingly, if health promotion research is the process through which knowledge on health-related behaviors is "discovered," and science equals knowledge, then health promotion research must be considered a science, right? To answer this question fully and accurately, we first provide an overview of what constitutes science and describe key concepts from its related discipline—the philosophy of science. Philosophy is from the Greek word *philosophia*, meaning love (*philo*) and the pursuit of knowledge (*sophia*). Philosophy of science, therefore, pertains to the structure, principles, and components of the scientific process—the framework lovingly used for the pursuit of knowledge.

Another important and related term used often when discussing the philosophy of science is *epistemology*. Epistemology is the branch of philosophy that studies knowledge and attempts to answer the basic question: what distinguishes true (adequate) knowledge from false (inadequate) knowledge? There are different ways of acquiring knowledge, some of which will be described in more detail later in this chapter. As a health promotion practitioner or researcher you may be called upon to evaluate different types of health information. Learning to distinguish valid health-related information from false information will be an important skill.

Illustration of Key Concepts

The goals of this chapter are to introduce the various epistemological positions taken in pursuit of knowledge, such as authority, tenacity, logic, and, of course, scientific inquiry. Once you have built a solid foundation in epistemology, you may

then be in the position to ascertain whether or not specific instances of health promotion research constitute science. Of course, you will also be able to ascertain whether other field's inquiries can be considered science, "junk science," or neither. Consequently, you will also acquire insight about how science generates knowledge.

Epistemology: Ways of Acquiring Knowledge

Given that science is a process in which we gather knowledge and organize that knowledge systematically, does the way or method in which the knowledge is collected determine whether or not the process can be called science? The way in which knowledge is generated determines whether or not it should be deemed science and therefore added to an overall scientific body of knowledge. Just because an effort is conducted "in the name of science" does not necessarily make it science. Second, although *science* literally means knowledge, in the modern sense it refers more to a specific process that adheres to standards and methods of generating knowledge. Thus, it matters greatly indeed, for there are many different ways in which we come by our knowledge regarding the world. Some ways in which we gather knowledge may contribute to science, whereas other ways may appear to be scientific, but, in reality, are not considered science in the technical sense. Regardless of the method, knowledge is conveyed (although it may vary in value). Thus, not all knowledge is created equal. It is most important to know the different ways or methods in which we accrue knowledge so that the quality of the knowledge can be judged appropriately and the implications for choosing one method over another can be determined.

In the following section, we describe various ways of knowing such as authority, tenacity (sometimes called tradition), and logic and reason, which are not considered as science in the modern sense, but nevertheless provide new ideas and suggestions. In addition, we describe scientific inquiry as a way of knowing. The former methods (authority, tenacity, and logic) are ways in which knowledge is derived, so they are important to know. By understanding these alternative epistemologies you will better understand science. Consequently, you may embrace scientific inquiry as the most important way of knowing. As you will see, these ways must be evaluated rather subjectively and contrast sharply with scientific inquiry as a way of knowing because science is able to provide both a body of knowledge and a specific method in which to evaluate that knowledge.

Unfortunately, in some instances, science may not have yet contributed to a specific research goal and thus may not be an available source of knowledge. For example, in the early 1980s before the HIV virus was discovered, researchers

did not know what caused the decline in the immune system condition (AIDS) or how it was being transmitted. The other ways of knowing (authority, tenacity, and logic) were the only sources of knowledge available. In essence, scientific inquiry is the only epistemology that requires data-driven conclusions. Other epistemologies exist regardless of whether data are available or whether people choose to consider the data. The challenge in those instances is to examine the underlying source of the knowledge and make the best determination regarding its accuracy.

Authority. Albert Einstein dubbed Galileo "the father of modern physics—indeed of modern science altogether" (Sobel, 1999, p. 326). Most everyone is familiar with the historical account of the renowned mathematician and scientist Galileo and his quest to advance the field of science, which up to that point in history was based mostly on the philosophy of Aristotle. Galileo sought to move beyond an understanding of why *phenomena* occur to how they occur. By phenomena, we mean events or circumstances evident to the senses and possible to describe scientifically. For Galileo, mathematics was one of his scientific tools:

> "Philosophy is written in this grand book the universe, which stands continually open to our gaze. But the book cannot be understood unless one first learns to comprehend the language and to read the alphabet in which it is composed. It is written in the language of mathematics, and its characters are triangles, circles, and other geometric figures, without which it is humanly impossible to understand a single word of it; without these, one wanders about in a dark labyrinth" [quoted in Drake, 1957, p. 237].

Although most scientists concur that Galileo's greatest contribution to science is his application of mathematics to the study of motion, he has been immortalized for his defense of Copernicus's theory that posited the sun rather than the Earth is the center of the cosmos. This sun-as-center-of-the-universe perspective was not only contrary to the Aristotelian perspective (that is, the Earth is the center of the cosmos), but more important, it contradicted the teachings and beliefs espoused by the Catholic Church. Interestingly, the Church also based its views on the Aristotelian perspective.

During this period in history, the Catholic Church was the moral and philosophical authority. If the Catholic Church said it was so, then most people accepted it for fact. Most people did not understand the order of the universe at this time. Thus, it was easier for people to defer to some authority, one who held some degree of knowledge regarding an issue. People during this era mostly obtained

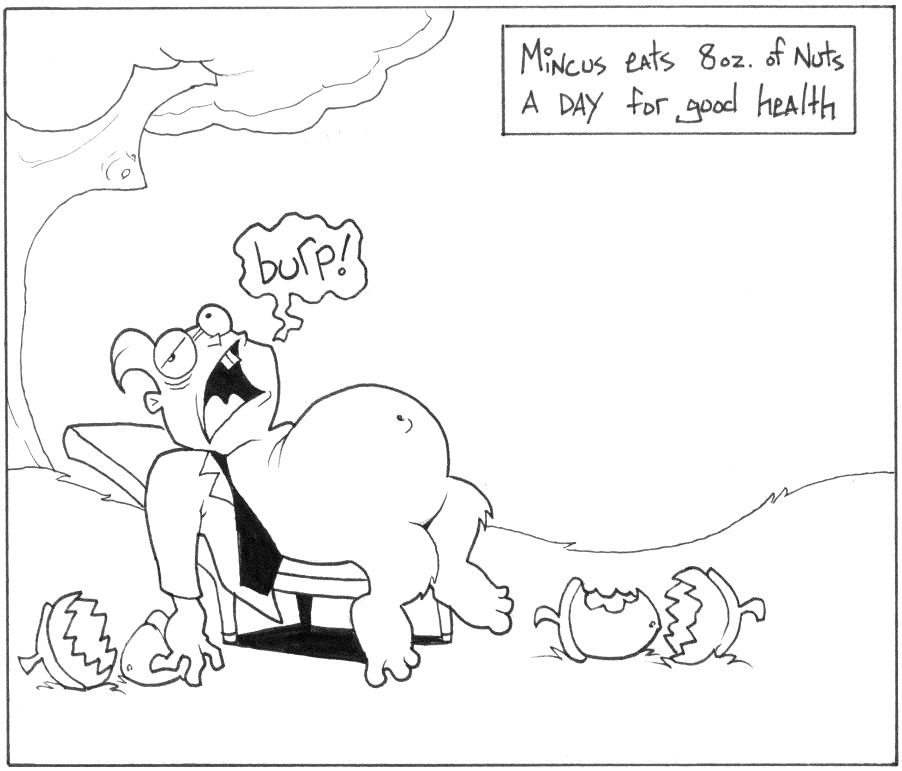
their knowledge through authority because there were not many other sources. This approach has a major shortcoming, however; the authority in question can be wrong, as was the case with the Church. Unfortunately, because Galileo went against the most powerful authority of his time, he was tried by the Church in front of the Inquisition. Because he would not recant his beliefs, he was sentenced to life imprisonment, which was later reduced to house arrest (Sobel, 1999).

In his search for knowledge Galileo truly believed that, "the authority of a thousand is not worth the humble reasoning of a single individual" (Redondi, 1987, p. 37). Even in our modern times, this quote should still apply. Yet, many of us still rely upon authority for facts and guidance. For example, many of us look up to our religious leaders as well as our politicians, physicians, and professors. Reliance on authority is not always problematic. Yet, not taking the time to evaluate the validity of an authority's ideas, suggestions, or comments may be problematic. Of course, an evaluation of the knowledge would require knowing what evidence (if any) the authority is using. For example, when a politician describes our current state of the union, we must consider whether or not the description is based on scientific evidence (such as gathering of key economic indicators, scientific surveys, archival data, and so on) or on anecdotal evidence from a few people, or, even worse, is not based on any evidence but is simply rhetoric. Furthermore, when a physician instructs someone to drink eight eight-ounce glasses of water per day, the person can ask if this recommendation is based on scientific evidence or if the recommendation is based on tenacity with no real basis (see Box 2.1, which describes an investigation into the origins of the health recommendation to drink eight glasses of water per day). Even with your professors (including the one teaching this course), you should question knowledge presented in class and inquire about the epistemology. These are only a few ways in which

Box 2.1. The "8 × 8" Recommendation

Valtin (2002) conducted a review of the scientific literature to determine the origin of the advice to "drink at least eight eight-ounce glasses of water per day." He found no scientific studies in support of the "8 × 8" health belief, and he could not find its origins in the scientific literature. In fact, he found evidence to suggest that such large amounts of water are not necessary in *healthy* adults, who live in temperate climates and are largely sedentary. Moreover, his review of a large body of related research suggested that the human body's osmoregulatory system maintains water balance naturally. Thus, it would appear that the ubiquitous "8 × 8" water recommendation is without scientific merit, but persists as a modern myth that is viewed as medically beneficial.

MINCUS TAKES DR. DINCUS'S ADVICE FOR GOOD HEALTH.



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to question authority, but as health promotion researchers and purveyors of knowledge, you should learn to question the knowledge that is presented to you. Moreover, you have an obligation to employ scientific inquiry regardless of the prevailing epistemologies. As with Galileo, however, your research may not be popular or readily accepted by society. Yet, as Albert Einstein stated, "A foolish faith in authority is the worst enemy of truth" (quoted in Calaprice, 2000, p. 303).

Tenacity or Tradition. Even in this postmodern age of technology, it is interesting that many health beliefs that have been around for a very long time and are sometimes referred to as "wives' tales" remain firmly entrenched in our minds and culture and continue to influence our health behaviors. For example, were you

told that it is dangerous to swim unless you wait an hour after eating? Were you scolded for going outside as a child with wet hair because it would cause you to catch a cold? Did your mother stress to you that you should eat all of your carrots because it will improve your eyesight? Or certainly you have heard the expression, "an apple a day helps keep the doctor away"? Do you still believe that chocolate causes acne? Are you familiar with the notion that oysters are an aphrodisiac? Many young women maintain that they cannot get pregnant while breastfeeding or the very first time they have vaginal intercourse. A more recent health belief that has become somewhat entrenched in modern society is that the flu vaccine will cause you to get the flu. Not surprising, some of these health beliefs may be rooted in truth, which may explain their longevity. For example, eating vegetables high in vitamin A (for example, carrots) is good for maintaining healthy eyesight; however, more than the recommended daily requirement will not improve your eyesight. What is surprising is that some of these health beliefs continue to persist despite the fact they have been proven false—hence the term *tenacity*. For example, many people still believe that going out in cold weather with wet hair will cause them to catch a cold, even though science has shown it is exposure to a virus that causes a cold.

Why do these beliefs persist even though they are false? Many of these beliefs have been handed down from one generation to the next through storytelling, through printed material, and presently through other forms of media. For many cultures, these beliefs become their traditions, and as traditions they tend to provide people with an acceptable way of exerting control over certain unavoidable events. Thus, accepting traditional beliefs as a way of knowing at a very basic level may serve to help people understand and exert control over their environment.

Health beliefs based on tradition, whether true or not, may also contribute to the cohesiveness of a cultural group. For example, unified cultures are grounded in their acceptance of similar beliefs and traditions. It is for these reasons that there may be problems with using only tenacity as a valid way of knowing: if the beliefs are erroneous, then there will be great difficulty in changing or discrediting them once they have been accepted widely and are entrenched within a culture. It may be useful for you to think about different health beliefs you have accepted and consider whether or not they are based on tradition, evidence, or truth.

Logic and Reasoning. The term *logic* is derived from the Greek word *logos*, which traditionally means "word," "thought," "principle," or "speech." *Logos* has been used among both philosophers and theologians and embodies both human reason (that is, intellect, capacity to discern and distinguish) and universal intelligence (that is, the Divine). Although philosophers are undoubtedly concerned with both

aspects of *logos,* as health promotion researchers we are concerned mainly with the human-reason aspect for this construct. Reason, along with its cousin term *logic,* is the foundation of philosophy and is still in use today as a way of knowing.

In modern terms, *logic* can be defined as the science of reasoning, proof, thinking, or inference. Logic is useful in that it allows you to analyze arguments or a piece of reasoning and determine whether or not they are accurate or "illogical." Although logic is considered a science, knowing the basics of logic and reasoning can assist you in spotting which arguments are invalid and which conclusions are false.

As previously stated, the Catholic Church based much of its beliefs and knowledge on another authority—Aristotle. Because Aristotle concerned himself with the investigation of natural phenomena, he was able to make many observations about the world. He then used logic and reasoning to define, analyze, and systematize his observations to make sense of what he observed. Logic and reasoning were his tools; thus, in the absence of science, logic and reasoning were the accepted standards of the time.

Specifically, Aristotle used an approach called a *syllogism,* which means an argument of a very specific form consisting of two premises and a conclusion, to amass his knowledge. Generating knowledge regarding the Earth and the cosmos, Aristotle used the following syllogism:

> Again, everything that moves with the circular movement, except the first sphere, is observed to be passed, and to move with more than one motion. The earth, then, also, whether it move about the centre or as stationary at it, must necessarily move with two motions. But if this were so, there would have to be passings and turnings of the fixed stars. Yet no such thing is observed. The same stars always rise and set in the same parts of the earth [Aristotle, 350 B.C. (2004), part 14, paragraph 1].

Using this syllogism, Aristotle concluded that "the earth, spherical in shape, is at rest in the centre of the universe" (Aristotle, 350 B.C. (2004), part 14, paragraph 4). Of course, it was later ascertained that Aristotle's conclusion was inaccurate. His syllogism contained a *fallacy.* In logic, the term *fallacy* has a very specific meaning: a fallacy is a technical flaw which makes an argument unsound or invalid. In this instance, the premises of Aristotle's argument were not true. Although his conclusion was sound, it was based on false premises and was therefore false. There are many other types of fallacies that occur quite frequently, such as a fallacy called *cum hoc ergo propter hoc* ("with this, therefore because of this"), which is to assert

that because two events occur together, they must be causally related. An example for this type of fallacy would be

> Teenagers eat a lot of chocolate.
>
> Teenagers have acne.
>
> Therefore, eating chocolate causes acne.

Another fallacy is called *converse accident* or *hasty generalization* and is the reverse of the fallacy of *accident*. The former occurs when you generalize to an entire group based on a few specific cases, which aren't representative of all possible cases. For example, "Professor Dincus is eccentric. Therefore all professors are eccentric."

The latter fallacy (accident) is also referred to as a sweeping generalization and occurs when a general rule is applied to a particular situation, but the features of that particular situation mean an exception to the rule should be made. For example, "College students generally like junk food. *You* are a college student, so you must like junk food." Finally, one last fallacy is called *post hoc ergo propter hoc* (after this, therefore because of this), which occurs when a cause-and-effect relationship is assumed because one factor occurred temporally before the other factor. For example, "After receiving his flu vaccine, Rick came down with the flu. Therefore, we must avoid the flu vaccine because it *caused* him to get the flu."

In present times, people still rely upon information that is derived from logic and reasoning, especially if there are not alternative ways of knowing. Yet, much information derived from logic and reasoning is based on arguments that may contain fallacies. Another issue with this way of knowing is that arguments cannot determine whether a statement is correct. Thus, as a way of knowing, logic and reasoning are useful but only if the arguments presented do not contain fallacies and if there are alternative ways to verify the conclusions. The challenge is to critique any arguments or information presented to you and to attempt to uncover any fallacies.

Scientific Inquiry as Epistemology

Scientific inquiry or research is conducted as a means to test ideas, to evaluate questions, and to determine how things work. Generally speaking, the goal is to generate knowledge regarding the nature of our universe. The nature of our universe can range from knowing and understanding weather patterns to knowing and understanding people's exercise patterns. Scientific inquiry is simply another way of knowing, but as you will discover, it is quite different from the other epistemologies that were described previously. For one thing, "research is a

disciplined way we come to know what we know" (Bouma and Atkinson, 1995, p. 9). It involves a process as described in Chapter One, but it also posits certain structures or components to that process. Concepts and structures such as *empiricism, data, theory, hypothetico-deductivism,* and *falsification* are the main building blocks of scientific inquiry and constitute the foundation of modern science. Furthermore, it is because of these concepts and structures that scientific inquiry as an epistemology is considered more reliable and valid than other methods of inquiry.

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In a Nutshell

Concepts and structures such as empiricism, data, theory, hypothetico-deductivism, *and* falsification *are the main building blocks of scientific inquiry and constitute the foundation of modern science.*

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Empiricism. From its inception, science began as a process that was viewed as objective and *empirical*. From the Greek word *emperirikós* meaning experienced or skilled, empirical is a concept that denotes observation and experimentation and is therefore linked literally to the concept of scientific inquiry. Moreover, the term *empiricism* represents the philosophical position that true knowledge is a product of sensory perceptions gleaned from observation. The strength of adopting this position is that many important research questions can be answered objectively through the collection of empirical *data*.

Data. Data are essentially concrete facts, records, or collections of information regarding some aspect of our universe that take on meaning when applied to a research question. Data are plural; datum is singular. For example, does Brand X cholesterol-lowering drug reduce LDL cholesterol more effectively than Brand Y drug? is a research question for which there is an answer that can be derived from empirical data. In this instance, empirical data collected to answer this question are the LDL cholesterol levels of people before taking the drugs and after taking the drugs. A limitation of adopting empiricism is that some questions cannot be answered in this manner because the answers cannot be derived from empirical data. For example, is Brand X cholesterol drug's name better than Brand Y cholesterol drug's name? is a question for which there is not empirical data available. Of course, we could modify the question into an empirical question by asking, do women aged forty-five to seventy-five who have high cholesterol judge

Brand X's name to be better than Brand Y's name? We could then survey women in this age category who have high cholesterol and ask them which name they prefer. The survey responses would constitute the empirical data. Thus, when conducting research to answer a health-related question, it is critical that the research question is constructed so that it is an empirical question from the beginning.

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In a Nutshell

Data are essentially concrete facts, records, or collections of information regarding some aspect of our universe that take on meaning when applied to a research question. Data are plural; datum is singular.

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Theory. If science were a movie, then theory would be the screenplay. No matter how talented the actors are, a good screenplay is still necessary to provide the framework in which the actors showcase their talents. Without the screenplay, they are simply improvising. For some talented actors, improvisation may work; however, for the majority of actors, a cohesive and intriguing plot coupled with realistic dialogue is critical. In science, theories play a major role as well because they provide an understanding of phenomena. In other words, they organize relationships among "characters" (that is, observations) into a coherent picture of what it all means. They provide a starting point for making future predictions as well. Furthermore, theories guide the research question.

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In a Nutshell

If science were a movie, then theory would be the screenplay.

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What exactly constitutes a scientific theory? How are theories generated? And how does a theory contrast with a hypothesis? Many people use the term *theory* interchangeably with *hypothesis,* and you will read instances in textbooks and research articles that use *theory* when they mean *hypothesis;* however, theory expands upon simple hypotheses and considers sets of relationships. *Hypotheses* are statements

that specify the nature of the relationships between variables, whereas a *theory* is much more involved and provides order and structure to sets of relationships. For example, "obesity is related to increased risk of breast cancer among postmenopausal women" is a hypothesis that could be tested by comparing incidence of breast cancer among postmenopausal women who are not obese to incidence among postmenopausal women who are obese. A theory, however, would account for differences in breast cancer rates by describing the causative interaction of physiological, environmental, and psychological variables.

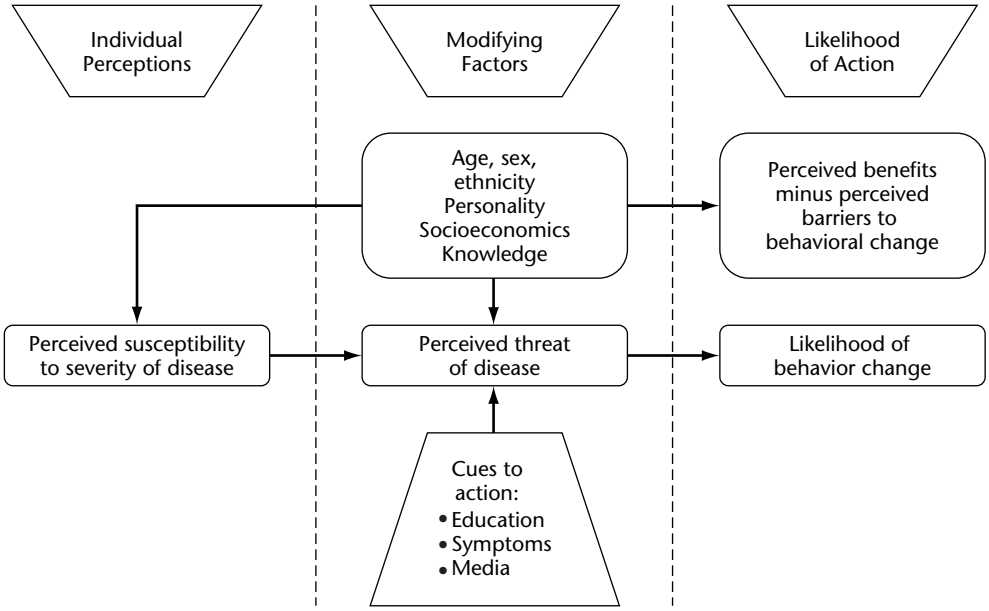
It is important to note that the phenomena involved in a theory must also be verifiable, meaning we can measure them or observe them either directly or indirectly. Once phenomena are measured or observed, understanding them translates into two types of theoretical explanations: *phenomenological* and *explanatory*. Phenomenological is when the phenomena are described and generalized, but without specific reference to causal mechanisms. In contrast, an explanatory theory identifies and explains the underling causal mechanisms.

Most theories used in health promotion research are phenomenological theories that specify which variables are involved in health-related behaviors and how the various variables interact to determine the behavior. For example, the Health Belief Model (HBM) is a theory of behavior developed in the 1950s by a group of social psychologists who wanted to better understand the widespread failure of people to participate in programs to prevent or to detect disease (Janz, Champion, and Strecher, 2002). The HBM is depicted in Figure 2.1 and posits that three major components, individual perceptions, modifying factors, and likelihood of action influence behavior. The HBM has been applied to the understanding of health behaviors where much research has generated support for the theory. Such behaviors include flu inoculations, breast self-examination, high blood pressure screening, seatbelt use, exercise, nutrition, smoking, and regular checkups. Other types of behavior that have been tested also include compliance with drug regimens, diabetic regimens, and weight loss regimens.

Given the critical role of theory in science, how are theories generated initially before they are widely implemented? In the case of the HBM, the theory grew out of the academic backgrounds of the theorists and the other theories to which they were exposed. Although there are several strategies for developing theories, such as intuition, previous knowledge, and personal observation, only two main strategies will be described in this section: *Baconian Inductivism* and *hypothetico-deductivism*. Both methods were conceptualized centuries ago and are still in use today. Each has its own strengths and weaknesses, but determining which method to use depends greatly on the domain or scope in which the theory will apply.

If Galileo was the father of modern science, Francis Bacon (1561–1626) was its champion. During the time Galileo was honing new methods to study

FIGURE 2.1. HEALTH BELIEF MODEL COMPONENTS AND LINKAGES.



phenomena, Francis Bacon was a philosopher (among other professions) who "proposed an entirely new system based on empirical and inductive principles and the active development of new arts and inventions, a system whose ultimate goal would be the production of practical knowledge for 'the use and benefit of men' and the relief of the human condition" (Internet Encylopedia of Philosophy, n.d., paragraph 1). His system or method was later called Baconian Inductivism, and it comprised four steps:

- Observation and classification of relevant facts
- Generalization by means of inductive reasoning
- Construction of a theoretical framework that allows one to deduce predictions from its laws and postulates
- Verification of the predictions

Although much improved over other nonscientific methods for generating knowledge, Baconian Inductivism has two major flaws: it cannot be used to derive theories regarding phenomena that are unobservable (for example, gravity), and its validity depends on a large number of observations under varying conditions.

Consequently, it did not emerge as an accepted standard for conducting scientific research and was not embraced widely. In certain instances, however, there may be no other alternative. For example, in the beginning stage of the U.S. AIDS epidemic, two physicians in Los Angeles observed that typically harmless opportunistic bacterial and viral infections among several young, gay male patients were making them extremely sick. Simultaneously, other similar observations were being noted among young, gay men in other cities. It was the emergence of these early anomalous observations that prompted scientists to convene and infer that a serious health issue was looming. From these early irregular observations general hypotheses emerged to explain these data. (See Box 2.2, which describes the various hypotheses that emerged from the data collected during the onset of the AIDS epidemic in the early 1980s.) Of course, many of the early hypotheses were flawed because they were based on a small number of observations and did not consider other varying conditions (e.g., observations from people who contracted the syndrome and were not gay).

Box 2.2. Early AIDS Hypotheses

In the beginning of the epidemic, AIDS was referred to as GRID—Gay-related immunodeficiency disease. The name was not changed to Acquired Immune Deficiency Syndrome until 1982. These are some of the early AIDS "theories" (which actually should be viewed as hypotheses) that were inferred from the available data (Garrett, 1994).

- A virus called "cytomegalovirus" or "CMV superinfection" is the root, where repeated exposures to the virus result in the deadly syndrome.
- Gay men have been exposed to too many microbes and have "microbial overload," causing the immune system to self-destruct.
- The recreation drug called "poppers" (amyl nitrites) plays a critical role in the deadly syndrome along with other gay lifestyle practices such as "fisting," "rimming," or steroid skin creams.
- AIDS is caused by some new variety of the HTLV virus, the virus that causes immune system disruptions and cancer.
- AIDS is an opportunistic infection causing disease only in persons who are already immuno-compromised from other microbes or conditions.
- AIDS is a new manifestation of the hepatitis B virus.
- AIDS is caused by an unknown contaminant of the hepatitis vaccine (early experimental trials of the vaccine had been conducted with gay men).
- AIDS is caused by African swine fever.

Following Bacon in the late 1600s, Sir Isaac Newton (1642–1727) developed a scientific method later termed "hypothetico-deductivism," which was essentially the opposite of Baconian Inductivism. His method entailed beginning with a hypothesis or statement of general principles and then subsequently using deduction to make predictions for a specific event. The predictions could be tested through empirical observation and verified. If the hypothesis is not verified then, it may be modified or rejected.

The strength of hypothetico-deductivism lies in the ability to test any hypothesis and confirm its validity through empirical observation. For example, using this method, Newton was able to test his theory of gravitation. Newton used his theory of gravitation to guide him in deducing specific predictions for which he was then able to make observations that supported or verified his theory. Thus, as a method, hypothetico-deductivism was much improved over inductivism and emerged as the new method for generating reliable and valid knowledge.

However, hypothetico-deductivisim is not without its problems. Research conducted with the main purpose of confirming a theory may, over time, become irrelevant as the theory is modified and perhaps rejected. Moreover, a set of data cannot irrevocably confirm or prove that a theory is valid because different theories can be supported by the same set of data.

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In a Nutshell

Research conducted with the main purpose of confirming a theory may, over time, become irrelevant as the theory is modified and perhaps rejected. Moreover, a set of data cannot irrevocably confirm or prove that a theory is valid because different theories can be supported by the same set of data.

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Falsification. Sir Karl Popper (1902–1994), considered one of the most influential twentieth-century philosophers of science, asserted that for a theory to be science, a necessary condition is that the theory consist of hypotheses that could be falsified. According to Popper, *falsification* is the best approach for testing scientific theories and contrasts sharply with the verification approach. The theory must be organized in a way such that its assertions can be refuted. This necessity for a scientific theory to be falsifiable is known as the *demarcation criterion*. Advocating the use of falsifiability as a scientific method to test theories, Popper therefore rejected inductivism, because falsification dictates that you must begin with falsifiable hypotheses before observations are made. Then, you collect data to refute

them—not verify them. As noted earlier, it is easy for the same set of data to support or verify many different theories simultaneously. Thus, falsification provides more rigor and confidence in the conclusions.

Integration with the Research Process

Integrating the research process with these underlying scientific principles will provide a better understanding of health phenomena. To accomplish this integration, health promotion research should undertake the nine steps delineated in Chapter One while interjecting scientific structure to those steps. For example, theory should be used to aide in the development of the research question (step 2) and in the choice of variables to be measured (step 5). The research questions should be empirical questions formulated from theory and falsifiable. Moreover, the research design should allow for the falsification of the proposed hypotheses (step 4). The variables chosen also have to allow for empirical measurement. And finally, data analysis (step 8) should go beyond whether or not the data confirm or support the hypotheses to include the falsification of hypotheses. If the research process incorporates these scientific principles, then health promotion as a field will continue to expand and garner the respect of a true scientific discipline.

Summary

Over the centuries, science has revolutionized its sources of knowledge, moving from authority, tenacity, and logic and reason to an elegant arrangement encompassing principles of empiricism, data, theory, deduction, and falsification. As science is considered a process that aids in our understanding of our world, throughout time, numerous scientific disciplines pertaining to specific aspects of our world have emerged. Health promotion can be considered one of the more modern scientific disciplines, but only if it adheres to sound scientific principles. In this chapter, we reviewed the philosophy of science from its origins to better understand the underlying concepts and the importance of science to health promotion research. We noted that one of the most critical scientific principles is the generation and rigorous testing of theory. As health promotion researchers investigating health-related behaviors, most of what we do should be grounded in theory. Theory constitutes the basic structure of our scientific inquiries for two reasons: theory guides the design of basic research studies and theory can be used to design a health promotion program that targets risk factors deemed modifiable and that are related to health-related outcomes.

Issues to Consider

1. In 2001, the Department of Health and Human Services assembled a panel of experts to determine whether condoms are effective in protecting against sexually transmitted diseases. Their conclusion was that the evidence was generally insufficient to support condom effectiveness. The Centers for Disease Control later retracted their fact sheet about condoms from their Website. What are the epistemologies that apply here?
2. Because behavior cannot be isolated (like a pathogen) to determine its causes, what are some scientific issues to consider when generating theories of behavior? Also, given that human behavior, especially health-related behaviors, is complex and multifaceted, how confident would you be in attributing a specific cause to a specific behavior such as smoking?
3. A theory in the scientific sense goes beyond simple hypotheses and provides order and structure to sets of relationships. A theory must also be verifiable either directly or indirectly. Given this definition of theory, what arguments would you present in debating whether or not Creationism should be considered a scientific theory taught in high school science classes?

For Practice and Discussion

You can get a better idea of the different ways we gather knowledge and the implications of choosing one way over another by the following hypothetical example. Say, for instance, that you have been given funds to design a program to prevent heart disease. Before you decide what type of program you would like to conceptualize and implement, you determine there are several possible ways in which you can make your decision. What are all of the different sources you can turn to that will help you conceptualize your program? Will you base your decision on what a cardiologist thinks is the best approach (use authority)? If you are to use logic and reason, what are your premises and conclusion? (for example, men who make more than $100,000 per year are Type A personalities; men with Type A personality will develop heart disease; therefore, men who make more than $100,000 per year are at risk for heart disease). Are there health beliefs related to heart disease that you already know are true? (for example, an apple a day keeps the doctor away). Which approach would you base your decision on and why? How would you examine the underlying sources of the nonscientific approaches?

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